

along a particular process line and you have the opportunity to apply automation. Right now, all those automation tools are on the cloud or on a local server, but is there some way or some use case where you see potential for the NPU to accelerate workloads over there?

Kedar: I think there isn't a single use case where the NPU will not shine or show its benefits, right? Because AI is going to be pervasive, there is no doubt about it. Whether you look at things like manufacturing, operations, health care, engineering, etc., you name the industry, and it's going to start to become more prevalent. The point is, with Qualcomm, we've always had a very heterogeneous architecture with our platforms. What that means is we have dedicated blocks for a lot of the cores like it's not just a CPU and GPU type platform. You have a CPU, a GPU, an NPU, you have an audio core, a video core, a camera core and so on. For example, if you're running a video-related task, we have a dedicated video call where it's going to do a lot of the offloading. Then we have for example, our audio block has a small little NPU inside that block, and so when you talk about noise suppression and a lot of these use cases, they run on these dedicated NPUs. So, it just depends on the use cases.

Even at Snapdragon Summit, where we announced the Snapdragon X Elite, we had several ISVs showcasing the benefits of what happens when you port use cases to Snapdragon. One of them, for example, was super resolution. Second was, you know, photo editing and it was shown by a company called Luminar Neo, in partnership with us. If you have a picture that is low in resolution and you want to optimize that picture, it does so in a matter of seconds and a significantly lower power relative to anything that you'd see from competition.

And so, it's just the beginning. I think you'll see almost every single use case getting optimized on the NPU, and obviously what that means is better performance, lower power, and translates into better battery life.

Digit: And based on the kind of use cases that you see growing, do you feel like you might have to go back to the design table for your next iteration and changes the amount of compute that has been given to, let's say, part of the SoC that's handling the audio, or maybe the part that handles floating point or integer instructions because that's usually a balance that keeps on changing. Or have you had requirements coming from one of your clients who believes that they have a certain amount of laptops that sell for a particularly use case, and that they like a bit more acceleration in that direction?

Alex: Well, I think what Kedar covered is the basis of the user interface actually changing into the device. As you as you can see, all of the all the functions that



Qualcomm Snapdragon X Elite

that we just talked about are running in the background. Now with LLMs and LVMs, visual models and Gen. AI, the user interface into the device is going to change and so the expectation from Microsoft is they're going to try to get the user from launching an app or typing or doing a search to actually talk into the machine. And having it do things for you so it can save time. And that whole concept when the new OS comes out is going to be compiled and optimized on top of Snapdragon, to begin with. And it's going to utilise the entire TOPS performance that we have on the NPU. And not only that, heterogeneously you can run stuff on the GPU or the CPU and you can share among all of these cores and then you get the best performance out of it. Of course, with the next iteration of our devices, we're going to increase those types of performances because

the more hardware acceleration that you make available, the more the software and the OS and the user experience is going to start to use that. And so now models are just on the device actually running in the background, and they're contextually aware of your screen. You can recall things you can. You can draw things and it'll generate an image for you and the reason why the NPU is important is because it goes through those accelerations relatively quickly with a lot lower power and and those functions can now be used over and over and over without draining your battery. So, forget the thin-and-light. All that stuff is table stakes. If you don't have that, you don't have a modern PC. But then if you don't have the AI functions we do, you don't have a modern AI PC. And so that's where everyone's excited about and how you interface to the machine is going to change how it saves time for you is going to change how it takes notes for you or completes your sentences or summarises a meeting or starts a presentation and builds maybe a word file. All those things can get done relatively easily with what we have to offer.

Digit: Now back in October, when you announced the Snapdragon X Elite, the performance stated was 45 tops for the NPU itself. Since there's going to be this eight-month difference from the time it was announced time it's being deployed, and considering optimization in this power curve, and all those things. Are you looking at a number higher than 45 by the time the silicon goes into production?

Alex: This generation is 45.

Kedar: This generation obviously is 45 tops, but one of the points which you'd hinted at earlier, the work happening on AI is changing drastically. There's a new model that comes out every day. And the scale that Qualcomm provides is the ability for us to provide sort of think of a horizontal layer where you have all of these different AI models that are rapidly coming out, talk about Lama or any of them that you know. A lot of the